Project: Student Feedback Analytics for Educational Institute

PART 5A — Model Selection

Based on the dataset (numerical input features → numerical output), we select **Linear Regression** to predict student satisfaction.

Goal: Predict **Overall Satisfaction Score** based on:

- Teaching Quality Rating
- Lab Facilities Rating

Model Type	Use Case	Why Selected / Rejected
Linear Regression	Supervised Prediction	✔ Selected — Simple and interpretable
Decision Tree	Regression / Classification	Alternative — may overfit small dataset
K-Means Clustering	Unsupervised grouping	Future scope for student segmentation
Logistic Regression	Classification	Not suitable for continuous satisfaction score

Step 1: Import Libraries & Load Data

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error,r2_score

df=pd.read_csv("student_feedback_cleaned.csv")

print(df.head())
```

Output :

```
             Timestamp                 Email address          Name
\
0  28/04/2026 13:38:31  bhansalitejas7498@gmail.com          Tejas
1  28/04/2026 13:38:39  yashsurana2005810@gmail.com    Yash surana
2  28/04/2026 13:38:58  bhansalitejas7498@gmail.com           Anit
3  28/04/2026 13:39:20  yashsurana2005810@gmail.com  Nirmal chatur
4  28/04/2026 13:39:27  bhansalitejas7498@gmail.com          Suraj

      Roll Number     Department        Year      Rate Teaching Quality
\
0                39       Computer  Third Year
3
1                22       Computer  Third Year
5
2                47       Computer  First Year
2
3                25       Computer  Third Year
4
4                60     Mechanical  Final Year
4

    Lab Facilities Rating     Faculty Support Rating    \
0                       4                          4
1                       4                          4
2                       4                          3
3                       5                          4
4                       5                          1

    Overall Satisfaction     Suggestions for Improvement    \
0                   Good
1                   Good                             No
2                Average                            NaN
3                   Good                             No
4                Average                            NaN

     Would You Recommend This Institute?
0                                    Yes
1                                    Yes
2                                     No
3                                    Yes
4                                     No
```

Step 2: Select Features and Target Variable

```
mapping={
"Excellent":3,
"Good":2,
"Average":1,
"Poor":0
}

df["Satisfaction_Code"]=df["Overall Satisfaction"].map(mapping)

# Features (X)
X=df[[
"Rate Teaching Quality",
"Lab Facilities Rating"
]]

# Target (y)
y=df["Satisfaction_Code"]

print("Features shape:",X.shape)
print("Target shape:",y.shape)
```

Output :
```
Features shape: (31, 2)
Target shape: (31,)
```

Step 3: Split Data into Training and Testing Sets (80:20)

```
X_train,X_test,y_train,y_test=train_test_split(
X,
y,
test_size=0.2,
random_state=42
)

print("Training samples:",len(X_train))
print("Testing samples:",len(X_test))
```

Output:
```
Training samples: 24
Testing samples: 7
```

Step 4: Train the Linear Regression Model

```
model=LinearRegression()

model.fit(X_train,y_train)

print("Intercept:",round(model.intercept_,2))

coeff_df=pd.DataFrame({
"Feature":X.columns,
"Coefficient":model.coef_
})

print(coeff_df)
```

Output:
```
Intercept: -0.01
                  Feature  Coefficient
0  Rate Teaching Quality     0.061768
1  Lab Facilities Rating     0.467119
```

Step 5: Evaluate the Model

```
y_pred=model.predict(X_test)

mse=mean_squared_error(y_test,y_pred)

rmse=np.sqrt(mse)

r2=r2_score(y_test,y_pred)

print(f"MSE : {mse:.2f}")
print(f"RMSE : {rmse:.2f}")
print(f"R2 Score : {r2:.4f}"
```

Output :
```
MSE : 0.44
RMSE : 0.67
R2 Score : 0.6490
```

Step 6: Sample Prediction

Predict satisfaction for:

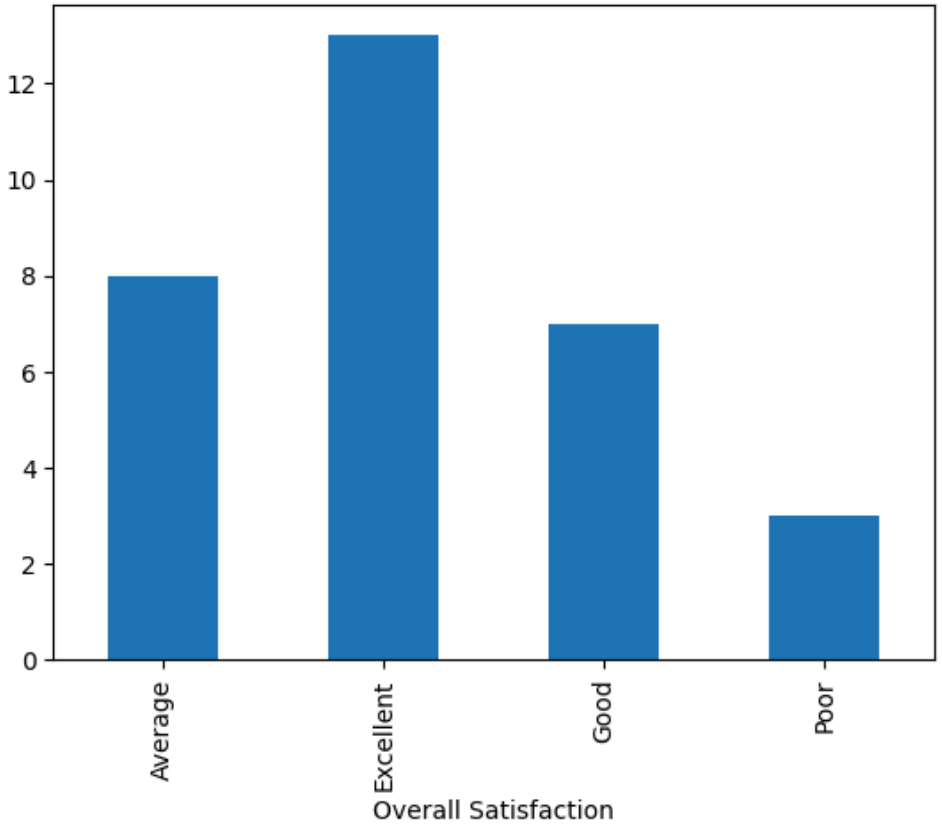
- Teaching Rating = 5
- Lab Rating = 4

```
new_student=[[5,4]]

predicted=model.predict(new_student)

print(
"Predicted Satisfaction Score:",
round(predicted[0],2)
)
```

Output : `Predicted Satisfaction Score: 2.17`

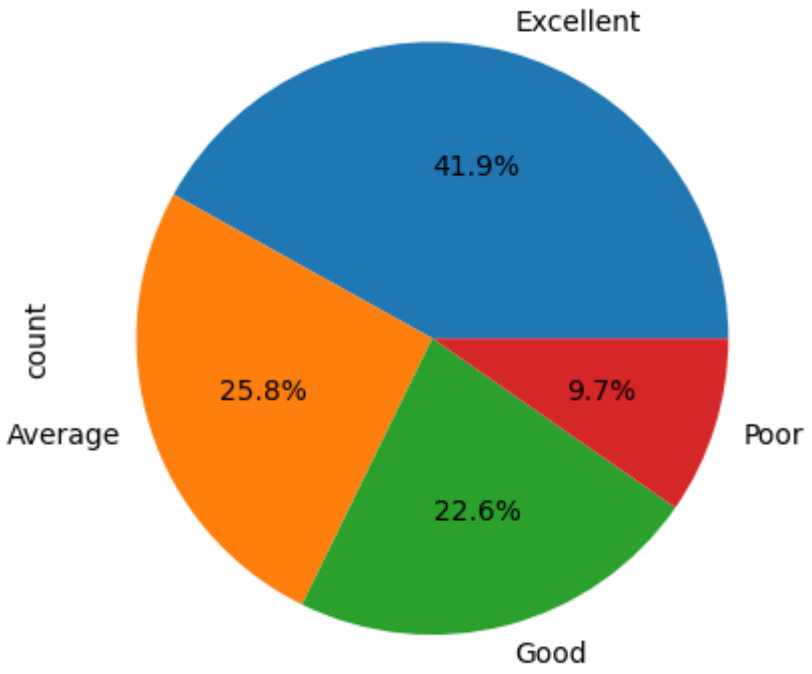
Bar Chart — Student Satisfaction Distribution

```
df.groupby("Overall Satisfaction").size().plot(
kind="bar"
)
```



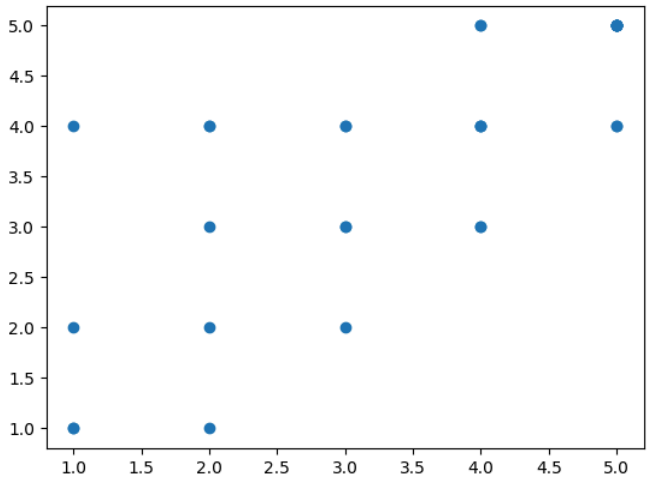
Pie Chart — Satisfaction Distribution

```
df["Overall Satisfaction"].value_counts().plot(
kind="pie",
autopct="%1.1f%%"
)
```



Scatter Plot — Teaching vs Lab Rating

```
plt.scatter(
df["Rate Teaching Quality"],
df["Lab Facilities Rating"]
)
```



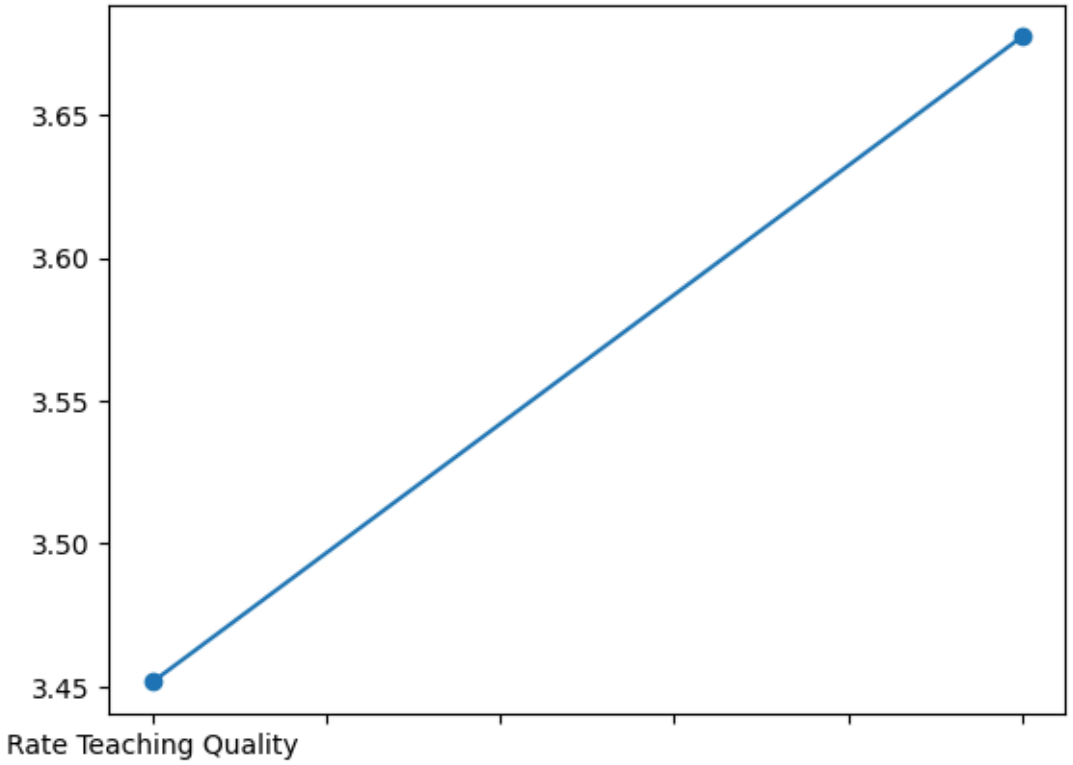
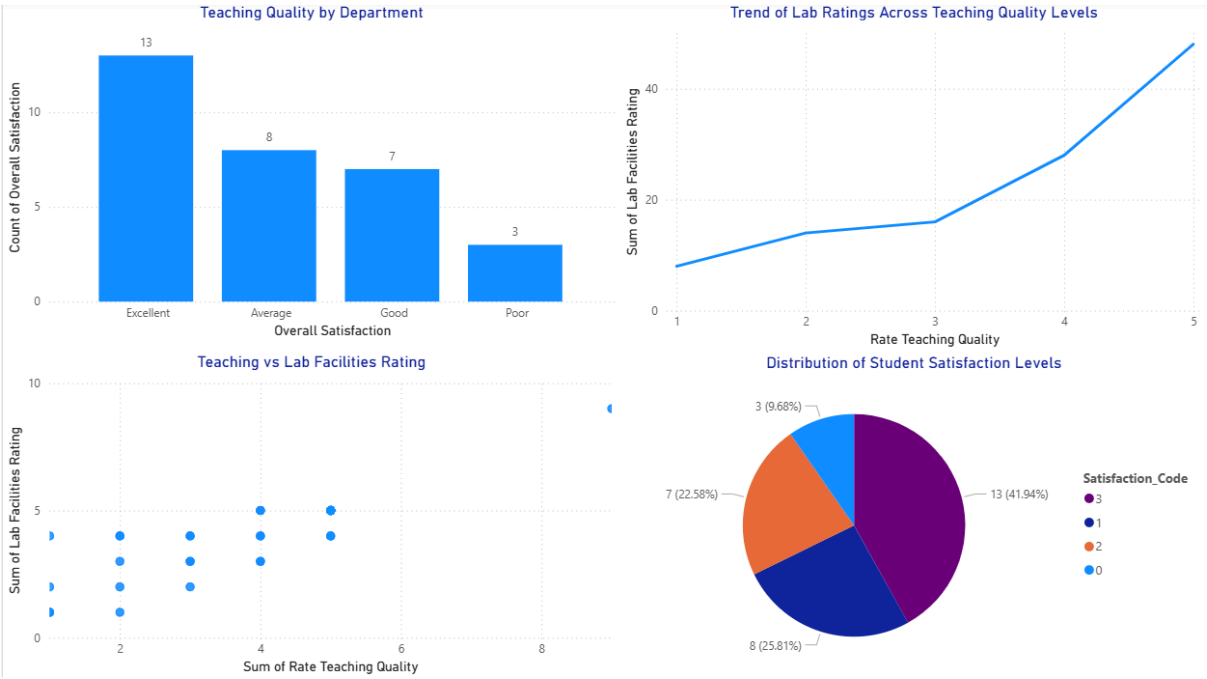
Heatmap — Feature Correlation *(optional extra)*

```
import seaborn as sns

sns.heatmap(
df.corr(numeric_only=True),
annot=True
)
```

Line Chart

```
ratings = df[[
"Rate Teaching Quality",
"Lab Facilities Rating"
]].mean()

ratings.plot(
kind="line",
marker="o"
)
```

Output:



Dashboard in Power BI